

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>
a) heat to convert and raise temperature of H₂O : mass of 1 L (1 0 0 0 g) of H₂O * specific heat capacity of H₂O * (6 0 - 3 0) C = 1 0 0 0 * 1 * 3 0 = 3 0 0 0 0 cal
b) heat to raise temperature of aluminum : mass of 1 kg (1 0 0 0 g) of aluminum * specific heat capacity of aluminum * (6 0 - 3 0) C = 1 0 0 0 * 0 . 2 1 5 * 3 0 = 6 4 5 0 cal
total heat required : 3 0 0 0 0 + 6 4 5 0 = 3 6 4 5 0 cal = 3 6 . 4 5 Kcal
</think>
<answer>
G . 3 0 Kcal , 6 . 4 5 Kcal
</answer>